

Quality Metrics for Quantifying Machine and Deep Learning Algorithms

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Outline

- ❑ Quality metrics for Performance evaluation of ML and DL Algorithms

Quality metrics for Regression Task

- Performance evaluation metrics of DL models for Regression Task :

- MAE
- MSE
- RMSE
- R2 Score

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}|$$

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y})^2$$

$$RMSE = \sqrt{MSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_i - \hat{y})^2}$$

$$R^2 = 1 - \frac{\sum (y_i - \hat{y})^2}{\sum (y_i - \bar{y})^2}$$

Where,

$\hat{y}$ – predicted value of y

$\bar{y}$ – mean value of y

Quality metrics for Classification Task

□ Performance evaluation metrics of DL models for Binary Classification Task :

- Image Based classification Task- Binary Data Classification
- Pixel based classification Task- Binary Data Segmentation

□ Performance evaluation metrics of DL models for Multiclass Classification Task:

- Image Based classification Task- Multiclass Data Classification
- Pixel based classification Task- Multiclass Data Segmentation

Quality metrics for Classification Task

- Performance evaluation metrics of DL models:
- Image 1: Segmented Image from algorithm/Method
- Image 2: Ground-truth Image (Manual Segmentation) or label Image
- Aerial Image :



- Segmented Output of Method



- Ground Truth (Label image)

Quality metrics for Classification Task

❑ *Confusion matrix for Binary Classification:*

		Actual	
		Positive	Negative
Predicted	Positive	True Positive	False Positive
	Negative	False Negative	True Negative

		Prediction	
		Positive	Negative
Actual	Positive	TP	FN
	Negative	FP	TN

Quality metrics for Classification Task

❑ Performance Evaluation metrics:

❑ True Positive (TP):

- It is an outcome where the model **correctly** predicts the **positive** class.

❑ True Negative (TN):

- It is an outcome where the model **correctly** predicts the **negative** class.

➤ False positive (FP):

- It is an outcome where the model **incorrectly** predicts the **positive** class.

❑ False Negative (FN):

- It is an outcome where the model **incorrectly** predicts the **negative** class.

Quality metrics for Classification Task

❑ **Confusion matrix for Multi-class Classification:**

❑ **3-Class classification**

		Predicted		
		Greyhound	Mastiff	Samoyed
Actual	Greyhound	P_{GG}	P_{MG}	P_{SG}
	Mastiff	P_{GM}	P_{MM}	P_{SM}
	Samoyed	P_{GS}	P_{MS}	P_{SS}

❑ **Iris Flower: 3-Class classification**

Total		Predicted		
		<i>Iris-setosa</i>	<i>Iris-versicolor</i>	<i>Iris-virginica</i>
Actual	<i>Iris-setosa</i>	12	1	1
	<i>Iris-versicolor</i>	0	16	0
	<i>Iris-virginica</i>	0	1	16

Quality metrics for Classification Task

- ❑ ***Confusion matrix for Multi-class Classification:***
- ❑ **n-Class classification**

		Prediction				
		Class 1	Class 2	Class 3	...	Class n
Actual	Class 1	Accurate				
	Class 2		Accurate			
	Class 3			Accurate		
	...				Accurate	
	Class n					Accurate

Quality metrics for Classification Task

- ❑ **Precision (Pr)** : Precision quantifies the ratio of pixels classified as object that are correctly identified.

$$Pr = \frac{TP}{TP + FP}$$

- ❑ **Sensitivity (Se) or Recall(Re) or TPR**: Sensitivity is how the algorithm is able to detect the object pixels.

$$Se = \frac{TP}{TP + FN}$$

- ❑ **Specificity (Sp) or TNR** : Specificity is how the algorithm is able to detect non- object pixels.

$$Sp = \frac{TN}{TN + FP}$$

Quality metrics for Classification Task

❑ Performance Evaluation metrics:

❑ True positive rate (TPR) : close to 1, better the model

❑ It expresses the fraction of pixels **correctly classified** as **true object pixels**.

❑ False positive rate (FPR): close to 0, better the model

❑ It expresses the fraction of pixels **wrongly classified** as **non-object pixels**.

❑ True negative rate (TNR): close to 1, better the model

❑ It expresses the fraction of pixels **correctly classified** as **non-object pixels**.

❑ False negative rate (FNR): close to 0, better the model

❑ It expresses the fraction of pixels **wrongly classified** as **true object pixels**.

Quality metrics for Classification Task

- With the help of these four values, we can calculate True Positive Rate (TPR), False Negative Rate (FNR), True Negative Rate (TNR), and False Negative Rate (FNR).

$$TPR = \frac{TP}{Actual\ Positive} = \frac{TP}{TP + FN}$$

$$FNR = \frac{FN}{Actual\ Positive} = \frac{FN}{TP + FN}$$

$$TNR = \frac{TN}{Actual\ Negative} = \frac{TN}{TN + FP}$$

$$FPR = \frac{FP}{Actual\ Negative} = \frac{FP}{TN + FP}$$

Quality metrics for Classification Task

- With the help of these four values, we can calculate , following metrics

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} * 100 \quad (3)$$

$$\text{Precision or } = \frac{TP}{TP + FP} * 100 \quad (4)$$

$$\text{Recall or TPR} = \frac{TP}{TP + FN} * 100 \quad (5)$$

$$\text{Specificity or TNR} = \frac{TN}{TN + FP} * 100 \quad (6)$$

$$\text{F1 Score} = 2 * \frac{\text{Recall} \times \text{Precision}}{\text{Recall} + \text{Precision}} * 100 \quad (7)$$

$$\text{False Positive Ratio(FPR)} = \frac{FP}{FP+FN} * 100 \quad (8)$$

Quality metrics for Classification Task

❑ EXAMPLE

- A machine learning model is trained to predict tumor in patients.
- The test dataset consists of 100 people.

		ACTUAL	
		Negative	Positive
PREDICTION	Negative	60	8
	Positive	22	10

Quality metrics for Classification Task

- **True Positive (TP)** — model correctly predicts the positive class (prediction and actual both are positive). In the above example, **10 people** who have tumors are predicted positively by the model.
- **True Negative (TN)** — model correctly predicts the negative class (prediction and actual both are negative). In the above example, **60 people** who don't have tumors are predicted negatively by the model.
- **False Positive (FP)** — model gives the wrong prediction of the negative class (predicted-positive, actual-negative). In the above example, **22 people** are predicted as positive of having a tumor, although they don't have a tumor. FP is also called a **TYPE I** error.
- **False Negative (FN)** — model wrongly predicts the positive class (predicted-negative, actual-positive). In the above example, **8 people** who have tumors are predicted as negative. FN is also called a **TYPE II** error.

Quality metrics for Classification Task

❑ Jaccard Index(JI) Score:

$$JI = \frac{TP}{TP + FP + FN} = \frac{|S \cap G|}{|S \cup G|} = \frac{|S \cap G|}{|S| + |G| - |S \cap G|}$$

❑ Jaccard Distance = 1- Jaccard Index (JI)

Where

S is the segmented output (Predicted) Image of the Algorithm/Model

G is the is the Ground Truth(Actual)or Target image

$|S \cap G|$ is the Intersection between S and G

$|S \cup G|$ is the Union between S and G

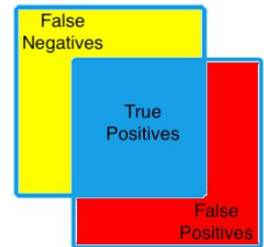
Quality metrics for Classification Task

Intersection of Union (IoU):

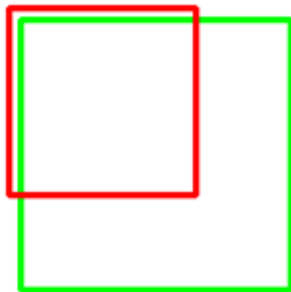
- It is used to measure the accuracy of an object detector on a particular dataset.
- PASCAL VOC 2012 challenge.

$$IoU = \frac{|S \cap G|}{|S \cup G|} = \frac{|S \cap G|}{|S| + |G| - |S \cap G|}$$

$$IoU = \frac{\text{Area of Overlap}}{\text{Area of Union}}$$

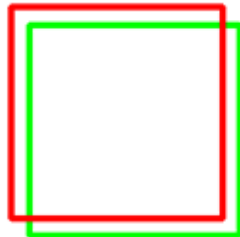


IoU: 0.4034



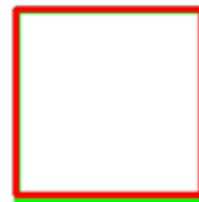
Poor

IoU: 0.7330



Good

IoU: 0.9264



Excellent



Quality metrics for Classification Task

❑ Dice Coefficient (DSC) or F1-score :

- ❑ The F1-score is the harmonic mean of precision (Pr) and recall (Re or Se) and it also has the property of better characterizing quality.

$$DSC \text{ or } F1 = \frac{2 * Pr * Re}{Pr + Re} = \frac{2TP}{2TP + FP + FN} = \frac{2 * |S \cap G|}{|S| + |G|}$$

- ❑ It achieves its maximum value of 1 when the segmentation of the positive class is perfect, and its lowest value of 0 when the segmentation is completely wrong.

❑ Dice Loss = 1- Dice Coefficient(DSC)

<https://towardsdatascience.com/common-classification-model-evaluation-metrics-2ba0a7a7436e>
<https://towardsdatascience.com/metrics-to-evaluate-your-semantic-segmentation-model-6bcb99639aa2>

Quality metrics for Classification Task

❑ Accuracy (Acc):

- ❑ The sum of this true positive (pixels which is correctly classified as object) and the true negative (pixels which is correctly classified as non-object and it is actually non object points) divided by the total number of pixel in the digital images is called accuracy.

$$Acc = \frac{TP + TN}{N}$$

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-
- Where, N is total number of pixels and $N = TP + TN + FP + FN$,

❑ Misclassification Rate:

- It tells you what fraction of predictions were incorrect.
- **It is also known as Classification Error.**
- You can calculate it using $(FP+FN)/(TP+TN+FP+FN)$ or ***(1-Accuracy)***.

Quality metrics for Classification Task

❑ Confusion matrix:

	Prediction is Positive	Prediction is Negative
Actual Outcome is Positive	True Positive (TP)	False Negative (FN)
Actual Outcome is Negative	False Positive (FP)	True Negative (TN)

- **Accuracy(Acc):** The ratio between correctly predicted outcomes and the sum of all predictions. $\text{Acc} = (TP + TN) / (TP + TN + FP + FN)$
- **Precision(Pr):** When the model predicted positive, was it right? All true positives divided by all positive predictions. $\text{Pr} = TP / (TP + FP)$
- **Recall (Re):** How many positives did the model identify out of all possible positives? True positives divided by all actual positives. $\text{Re} = TP / (TP + FN)$
- **F1-score:** This is the weighted average of precision and recall. $(2 \times \text{recall} \times \text{precision} / (\text{recall} + \text{precision})) = (2TP) / (2TP + FP + FN)$
- ❑ **IoU (Jl) score** = $TP / (TP + FP + FN)$

Illustration of Quality metrics for Classification Task

- $y_pred = [0.99, 0, 0, 0.99, 0, 0, 0.99, 0.99, 0.99, 0, 0.99, 0.99, 0, 0.99, 0.99]$ #Number of 0's=6 and Number of 1's=9

- $Y_true = [1, 1, 0, 1, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 1]$ #Number of 0's=8 and Number of 1's=7

- **# Analytical methods**

- **#TP:** It is an outcome where the model correctly predicts the positive class.

- **#TN:** It is an outcome where the model correctly predicts the negative class.

- **#FP:** It is an outcome where the model incorrectly predicts the positive class

- **#FN:** It is an outcome where the model incorrectly predicts the negative class

- **Hence, $TP(1)=6, FP(1)=3, TN(0)=5, FN(0)=1$**

- **## Class 1:**

- **#Precision(Pr): $TP/(TP+FP)=6/(6+3)=0.6666$;**

- **# Recall(Re): $TP/(TP+FN)=6/(6+1)=0.857$;**

- **# F1 score= $2*Pr*Re/(Pr + Re)=0.75$**

- **## Class 0:**

- **#Precision: $TN/(TN+FN)=5/(5+1)=0.8333$;**

- **# Recall: $TN/(TN+FP)=5/(5+3)=0.625$;**

- **# F1 score= $2*Pr*Re/ (Pr + Re)=0.714$**

- **#Overall or Average Accuracy: $(TP+TN)/(TP+TN+FP+FN) = 11/15=0.73$**

Illustration of Quality metrics for Classification Task

- **# SCIKIT Library**
- **# Python script for confusion matrix**
- **from sklearn.metrics import confusion_matrix**
- **from sklearn.metrics import accuracy_score**
- **from sklearn.metrics import f1_score**
- **from sklearn.metrics import precision_score**
- **from sklearn.metrics import recall_score**
- **from sklearn.metrics import classification_report**
- **predicted = [0.99, 0, 0, 0.99, 0, 0, 0.99, 0.99, 0.99, 0, 0.99, 0.99, 0, 0.99, 0.99] #Number of 0's=6 and Number of 1's=9**
- **actual = [1, 1, 0, 1, 0, 0, 1, 0, 0, 0, 1, 1, 0, 0, 1] #Number of 0's=8 and Number of 1's=7**
- **results = confusion_matrix(actual, predicted)**
- **accu_predicted=accuracy_score(actual, predicted)**
- **precison=precision_score(actual, predicted, average=None)**
- **recall=recall_score(actual, predicted,average=None)**
- **F1=f1_score(actual, predicted,average=None)**
- **report=classification_report(actual, predicted)**
- **Accuracy Report :**

	precision	recall	f1-score	support
Class 0	0.83	0.62	0.71	8
Class 1	0.67	0.86	0.75	7
accuracy			0.73	15
macro avg	0.75	0.74	0.73	15
weighted avg	0.76	0.73	0.73	15

Illustration of Quality metrics for Classification Task

- Macro F1-Score: we need to compute Macro-Precision and Macro-Recall before. They are respectively calculated by taking the average precision for each predicted class and the average recall for each actual class.

$$MacroAveragePrecision = \frac{\sum_{k=1}^K Precision_k}{K}$$

$$MacroAverageRecall = \frac{\sum_{k=1}^K Recall_k}{K}$$

Eventually, Macro F1-Score is the harmonic mean of Macro-Precision and Macro-Recall:

$$Macro\ F1-Score = 2 * \left(\frac{MacroAveragePrecision * MacroAverageRecall}{MacroAveragePrecision^{-1} + MacroAverageRecall^{-1}} \right)$$

Illustration of Quality metrics for Classification Task

- Micro F1-Score: we need to compute Micro-Precision and Micro-Recall before.

$$\text{Micro Average Precision} = \frac{\sum_{k=1}^K TP_k}{\sum_{k=1}^K \text{Total Column}_k} = \frac{\sum_{k=1}^K TP_k}{\text{Grand Total}}$$

$$\text{Micro Average Recall} = \frac{\sum_{k=1}^K TP_k}{\sum_{k=1}^K \text{Total Row}_k} = \frac{\sum_{k=1}^K TP_k}{\text{Grand Total}}$$

$$\text{Micro Average F1} = \frac{\sum_{k=1}^K TP_k}{\text{Grand Total}}$$

Illustration of Quality metrics for Classification Task

□ Example of Overall Accuracy (OA)- Total Accuracy

	Class types determined from reference source				
Class types determined from classified map	# Plots	Conifer	Hardwood	Water	Totals
	Conifer	50	5	2	57
	Hardwood	14	13	0	27
	Water	3	5	8	16
	Totals	67	23	10	100

$$Accuracy_{Total} = \frac{50 + 13 + 8}{100} * 100 = 71\%$$

Diagonals represent sites classified correctly according to reference data

Off-diagonals were misclassified

Illustration of Quality metrics for Classification Task

❑ Class-wise Accuracy (CA): User Accuracy

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	Class types determined from reference source				
Class types determined from classified map	# Plots	Conifer	Hardwood	Water	Totals
	Conifer	50	5	2	57
	Hardwood	14	13	0	27
	Water	3	5	8	16
	Totals	67	23	10	100

Example: Conifer

$$Accuracy_{User's, Conifer} = \frac{50}{57} * 100 = 88\%$$

Illustration of Quality metrics for Classification Task

❑ Example of Accuracy Assessment:

	Class types determined from reference source					
Class types determined from classified map	# Plots	Conifer	Hardwood	Water	Totals	User's Accuracy
	Conifer	50	5	2	57	88%
	Hardwood	14	13	0	27	48%
	Water	3	5	8	16	50%
	Totals	67	23	10	100	
Producer's Accuracy		75%	57%	80%		Total: 71%

Illustration of Quality metrics for Classification

Task

□ Kappa Coefficient (Ka):

- Another accuracy indicator is the **kappa coefficient**. It is a measure of how closely the classification results (instances classified) by the machine learning classifier matched the data labeled as ground truth.
- It can take **values** from 0 to 1. If **kappa coefficient** equals to 0, there is no agreement between the classified image and the reference image.

$$Ka = \frac{OA - \text{ChanceAgreement}}{1 - \text{ChanceAgreement}}$$

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- Observed accuracy determined by diagonal in CM
- Chance agreement incorporates off-diagonal
- – Sum of Product of row and column totals for each class. Where, N is total number of pixels and
- Mathematically

$$Ka = \frac{N \sum_{i=1}^k x_{ii} - \sum_{i=1}^k x_{i+} x_{+i}}{N^2 - \sum_{i=1}^k x_{i+} x_{+i}}$$

Illustration of Quality metrics for Classification

Task

□ Example of Kappa Coefficient:

	Class types determined from reference source					
Class types determined from classified map	# Plots	Conifer	Hardwood	Water	Totals	User's Accuracy
	Conifer	50	5	2	57	88%
	Hardwood	14	13	0	27	48%
	Water	3	5	8	16	50%
	Totals	67	23	10	100	
Producer's Accuracy		75%	57%	80%		Total: 71%

$$K_a = \frac{100 \times 71 - (57 \times 67 + 27 \times 23 + 16 \times 10)}{(100 \times 100) - (57 \times 67 + 27 \times 23 + 16 \times 10)} = 0.46$$

Quality metrics for Segmentation methods

□ Accuracy Assessment:

Classification results

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1686	Grass	Alfalfa	Cotton	Chili	Fallow (corn)	total	User accuracy (%)
Grass	110	22	0	0	0	132	83.3
Alfalfa	5	105	0	0	0	110	79.5
Cotton	0	0	945	5	0	950	99.5
Chili	0	0	50	42	0	92	45.7
Fallow	0	0	0	0	484	484	100
total	115	127	995	47	484	1768	
Producer accuracy (%)	95.6	82.7	95.0	89.4	100		

$$Overlay_Accuracy = \frac{1686}{1768} = 95.4\%$$

$$Kappa_Index = \frac{1686 - (132 \times 115 + 110 \times 127 + 950 \times 995 + 92 \times 47 + 484 \times 484) / 1768}{1768 - (132 \times 115 + 110 \times 127 + 950 \times 995 + 92 \times 47 + 484 \times 484) / 1768} = 89.3\%$$

Thank You